

JAVA PROGRAMMING LABORATORY
INDUSTRY PROBLEM-SOLVING TASK

Java AWT, Applets, Event Handling and Layout Managers

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Faculty	Dr. C. Sivasankar	Duration	Lab Session

INDUSTRY TASK 1: Employee Performance & Payroll Dashboard

Problem Statement: A software company wants to develop a simple Employee Performance and Payroll Management System using Java AWT. The application should allow the HR team to enter employee details, process salary information, and generate a structured payroll report.

Functional Requirements

- Develop an AWT-based graphical user interface.
- Accept Employee ID, Employee Name, Department, Designation, Basic Salary, Working Days, and Performance Rating.
- Use suitable AWT components such as Label, TextField, Choice, Checkbox/CheckboxGroup, Button, TextArea, and List.
- Calculate Gross Salary, Performance Bonus, and Net Salary using suitable business rules defined by the student.
- Display the complete employee payroll report in a List or TextArea when Generate Report is clicked.
- Use GridBagLayout or BorderLayout for arranging the interface.
- Implement button events using ActionListener.
- Provide Submit/Generate Report and Reset functionality.
- Validate important inputs such as empty fields and numeric salary values.

Industry Context: The application simulates an HR payroll module used to capture employee information, perform basic payroll processing, and present a report for HR verification.

Expected Deliverables

- Java source code.
- Working GUI application with event handling.

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- Sample input and output.
- Screenshot of the final application.
- Brief explanation of components, layout manager, and event handling used.

Java Code:

```
import java.awt.*;
import java.awt.event.*;

public class EmployeePayroll extends Frame implements ActionListener {
    Label title, idLabel, nameLabel, deptLabel, desigLabel;
    Label salaryLabel, daysLabel, ratingLabel;
    TextField idField, nameField, salaryField, daysField;
    Choice deptChoice, desigChoice;
    CheckboxGroup ratingGroup;
    Checkbox excellent, good, average, poor;
    Button generateButton, resetButton;
    TextArea reportArea;

    public EmployeePayroll() {
        setTitle("Employee Performance & Payroll Dashboard");
        setSize(700, 600);
        setLayout(new BorderLayout(10, 10));

        title = new Label("EMPLOYEE PERFORMANCE & PAYROLL DASHBOARD", Label.CENTER);
        title.setFont(new Font("Arial", Font.BOLD, 20));
        add(title, BorderLayout.NORTH);

        Panel formPanel = new Panel(new GridBagLayout());
        GridBagConstraints gbc = new GridBagConstraints();
        gbc.insets = new Insets(6, 6, 6, 6);
        gbc.fill = GridBagConstraints.HORIZONTAL;

        idLabel = new Label("Employee ID:");
        idField = new TextField(20);
        addComponent(formPanel, idLabel, gbc, 0, 0);
        addComponent(formPanel, idField, gbc, 1, 0);

        nameLabel = new Label("Employee Name:");
        nameField = new TextField(20);
```

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```
addComponent(formPanel, nameLabel, gbc, 0, 1);
addComponent(formPanel, nameField, gbc, 1, 1);
deptLabel = new Label("Department:");
deptChoice = new Choice();
deptChoice.add("CSE");
deptChoice.add("IT");
deptChoice.add("HR");
deptChoice.add("Finance");
deptChoice.add("Marketing");
addComponent(formPanel, deptLabel, gbc, 0, 2);
addComponent(formPanel, deptChoice, gbc, 1, 2);
desigLabel = new Label("Designation:");
desigChoice = new Choice();
desigChoice.add("Software Engineer");
desigChoice.add("Senior Engineer");
desigChoice.add("Team Lead");
desigChoice.add("Manager");
addComponent(formPanel, desigLabel, gbc, 0, 3);
addComponent(formPanel, desigChoice, gbc, 1, 3);
salaryLabel = new Label("Basic Salary:");
salaryField = new TextField(20);
addComponent(formPanel, salaryLabel, gbc, 0, 4);
addComponent(formPanel, salaryField, gbc, 1, 4);
daysLabel = new Label("Working Days:");
daysField = new TextField(20);

addComponent(formPanel, daysLabel, gbc, 0, 5);
addComponent(formPanel, daysField, gbc, 1, 5);

// Performance Rating
ratingLabel = new Label("Performance Rating:");

ratingGroup = new CheckboxGroup();
```

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```
excellent = new Checkbox("Excellent (5)", ratingGroup, false);
good = new Checkbox("Good (4)", ratingGroup, false);
average = new Checkbox("Average (3)", ratingGroup, true);
poor = new Checkbox("Poor (1-2)", ratingGroup, false);

Panel ratingPanel = new Panel(new FlowLayout(FlowLayout.LEFT));
ratingPanel.add(excellent);
ratingPanel.add(good);
ratingPanel.add(average);
ratingPanel.add(poor);

addComponent(formPanel, ratingLabel, gbc, 0, 6);
addComponent(formPanel, ratingPanel, gbc, 1, 6);

add(formPanel, BorderLayout.CENTER);

// Bottom section
Panel bottomPanel = new Panel(new BorderLayout());

Panel buttonPanel = new Panel();

generateButton = new Button("Generate Report");
resetButton = new Button("Reset");

buttonPanel.add(generateButton);
buttonPanel.add(resetButton);

generateButton.addActionListener(this);
resetButton.addActionListener(this);

reportArea = new TextArea(10, 60);
reportArea.setEditable(false);
```

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```
bottomPanel.add(buttonPanel, BorderLayout.NORTH);
bottomPanel.add(reportArea, BorderLayout.CENTER);
```

```
add(bottomPanel, BorderLayout.SOUTH);
```

```
// Window closing
```

```
addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        dispose();
    }
});
```

```
setVisible(true);
}
```

```
// Helper method for GridBagLayout
```

```
private void addComponent(Panel panel, Component component,
    GridBagConstraints gbc, int x, int y) {
```

```
    gbc.gridx = x;
    gbc.gridy = y;
```

```
    panel.add(component, gbc);
}
```

```
@Override
```

```
public void actionPerformed(ActionEvent e) {
```

```
    if (e.getSource() == generateButton) {
```

```
        String id = idField.getText().trim();
```

```
        String name = nameField.getText().trim();
```

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```
String salaryText = salaryField.getText().trim();
String daysText = daysField.getText().trim();

// Validation
if (id.isEmpty() || name.isEmpty() ||
salaryText.isEmpty() || daysText.isEmpty()) {

reportArea.setText("ERROR: Please fill all required fields.");
return;
}

try {

double basicSalary = Double.parseDouble(salaryText);
int workingDays = Integer.parseInt(daysText);

if (basicSalary <= 0) {
reportArea.setText("ERROR: Salary must be greater than 0.");
return;
}

if (workingDays < 1 || workingDays > 31) {
reportArea.setText(
"ERROR: Working days must be between 1 and 31.");
return;
}

// Salary calculation
double hra = basicSalary * 0.20;
double da = basicSalary * 0.10;

double grossSalary = basicSalary + hra + da;
```

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```
double bonusRate = 0;
String rating = "";

if (excellent.getState()) {
    bonusRate = 0.15;
    rating = "Excellent";
}
else if (good.getState()) {
    bonusRate = 0.10;
    rating = "Good";
}
else if (average.getState()) {
    bonusRate = 0.05;
    rating = "Average";
}
else {
    bonusRate = 0;
    rating = "Poor";
}

double performanceBonus = basicSalary * bonusRate;
double netSalary = grossSalary + performanceBonus;

// Report
reportArea.setText("");

reportArea.append("===== EMPLOYEE PAYROLL REPORT =====\n\n");
reportArea.append("Employee ID    : " + id + "\n");
reportArea.append("Employee Name  : " + name + "\n");
reportArea.append("Department    : " +
deptChoice.getSelectedItem() + "\n");
reportArea.append("Designation    : " +
desigChoice.getSelectedItem() + "\n");
```

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```
reportArea.append("Working Days    : " + workingDays + "\n");
reportArea.append("Performance    : " + rating + "\n\n");

reportArea.append("Basic Salary    : Rs. " +String.format("%.2f", basicSalary) + "\n");
reportArea.append("HRA (20%)      : Rs. " +String.format("%.2f", hra) + "\n");

reportArea.append("DA (10%)       : Rs. " +String.format("%.2f", da) + "\n");

reportArea.append("Gross Salary   : Rs. " +String.format("%.2f", grossSalary) + "\n");

reportArea.append("Performance Bonus : Rs. " +String.format("%.2f", performanceBonus) + "\n");

reportArea.append("-----\n");

reportArea.append("NET SALARY      : Rs. " +String.format("%.2f", netSalary) + "\n");

reportArea.append("\n=====");

} catch (NumberFormatException ex) {

reportArea.setText("ERROR: Salary and Working Days must be numeric.");
}
}

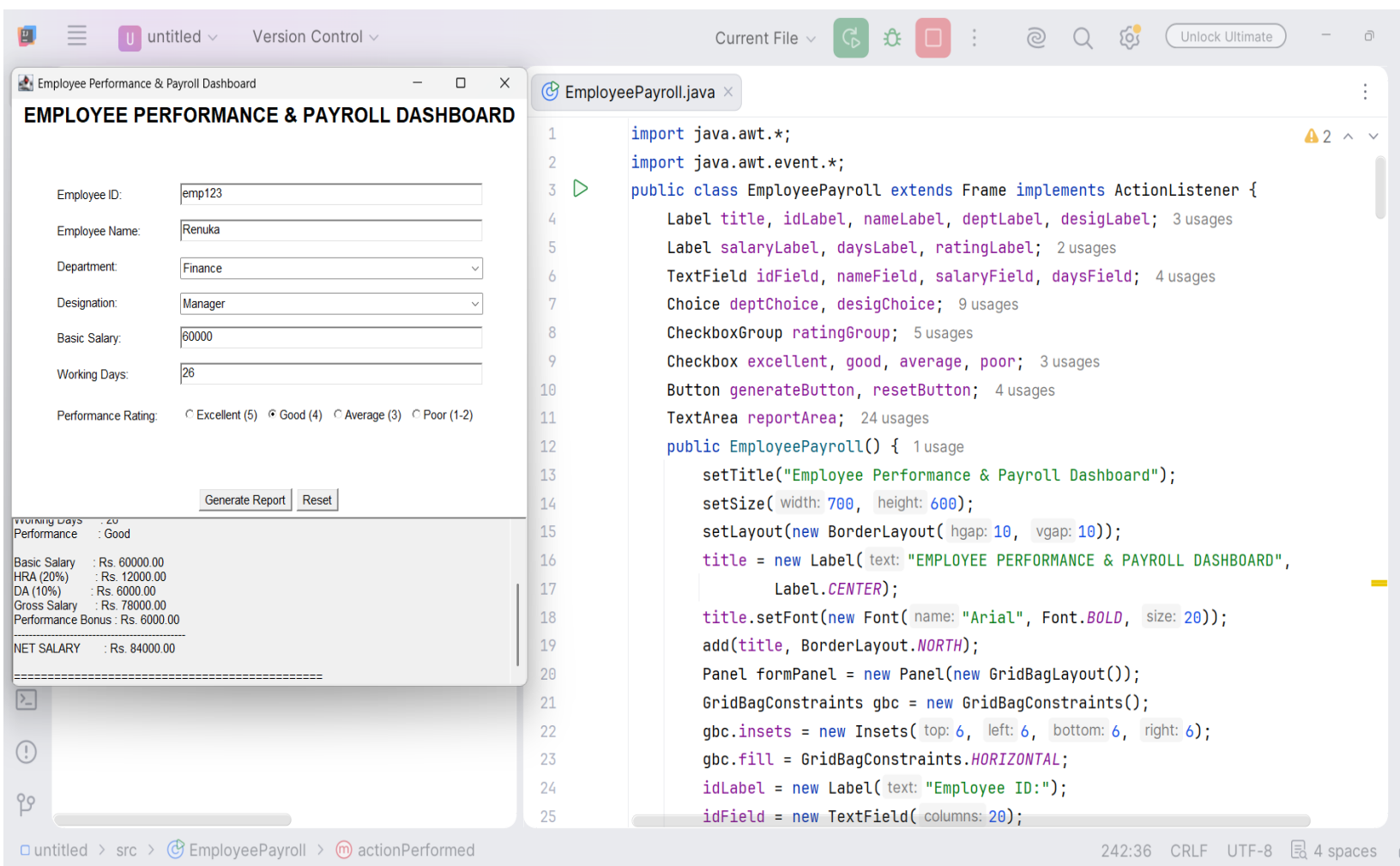
else if (e.getSource() == resetButton) {

idField.setText("");
nameField.setText("");
salaryField.setText("");
daysField.setText("");

deptChoice.select(0);
desigChoice.select(0);
```


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```
average.setState(true);
reportArea.setText("");
}
}
public static void main(String[] args) {
    new EmployeePayroll();
}
}
```



Employee Performance & Payroll Dashboard

AWT Components Used:

Component	Purpose
Label	Displays labels such as Employee ID, Name, Department, Salary, etc.

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TextField	Accepts Employee ID, Name, Basic Salary, and Working Days.
Choice	Allows selection of Department and Designation.
CheckboxGroup	Allows the user to select one Performance Rating.
Button	Provides Generate Report and Reset operations.
TextArea	Displays the complete payroll report.

Layout Manager:

GridBagLayout is used to arrange the employee input components systematically in rows and columns. BorderLayout is used to organize the main sections of the dashboard.

Event Handling:

The application implements ActionListener to handle button-click events. When **Generate Report** is clicked, the program validates the input, calculates Gross Salary, Performance Bonus, and Net Salary, and displays the payroll report. The **Reset** button clears all entered data.